

24-Mac Mini Game Development Cluster

24 dedicated machines · 24 parallel game projects · WineJS + KasmVNC · One rack · One workflow

1. Overview — What This Cluster Does

This is a **24-node Mac Mini cluster** designed for **parallel game development**. Each Mac Mini runs a dedicated game project — original game + target engine + map editor + tools — all accessible from a single browser on your ViewFinity screen.

Each machine runs **WineJS** (Windows apps in browser via KasmVNC + Wine), allowing you to run Windows-based game development tools (Hammer, WorldEdit, 3ds Max, etc.) and the games themselves — all in your browser, all persistent, all 24/7.

 **The Goal:** Remove all friction between inspiration and action. Watch a video → get inspired → click a project → all tools open exactly where you left them. No setup. No waiting. No remembering. Just work.

2. Hardware — 24 Mac Mini M4

The cluster consists of **24 Apple Mac Mini M4** units. These are chosen for their exceptional price-to-performance ratio, ultra-low power consumption, and silent operation — perfect for a 24/7 game development lab.

Component	Specification	Why It Matters
Model	Apple Mac Mini M4 (2024)	Latest generation, best performance per watt
Chip	Apple M4 (10-core CPU, 10-core GPU)	Integrated GPU with unified memory — no separate graphics card needed
Memory (RAM)	16GB unified memory	Sufficient for WineJS + game + editor. GPU can access all RAM
Storage	256GB SSD	Enough for OS + WineJS + one game project per machine
Thunderbolt	3x Thunderbolt 4 ports	High-speed interconnect for daisy-chaining nodes
Ethernet	Gigabit Ethernet	For management, SSH, and shared storage access
Power Supply	Built-in 155W (actual draw: 4-45W)	Extremely efficient — runs cool and quiet
Cooling	Active fan (silent under load)	No loud server fans — office-friendly
OS	Ubuntu 22.04 LTS (ARM64)	Free, lightweight, Docker-native, headless management

3. Rack & Physical Setup

The 24 Mac Minis are mounted in a **standard 19-inch server rack** using a **3D-printed scalable rackmount stand**. This design allows for dense stacking with proper airflow between units.

Rack Configuration

4 shelves · 6 units per shelf

24 Mac Minis total

Standard 19" rack width

3D printed brackets (STL available on GitHub)

Power Distribution

4x 12-socket PDU power strips

Each PDU handles 6 Mac Minis

Total power: ~960W at full load

Runs on a single UK 13A socket (3,120W capacity)

Networking

24-port Gigabit switch

For management, SSH, NFS shared storage

Thunderbolt daisy-chain for high-speed interconnect

Rackmount Design

Each Mac Mini sits on a 3D-printed bracket

Spacing between units allows airflow

Brackets attach to standard rack shelves

[GitHub: scalable-Mac-mini-stand](#)

4. Network Topology

The cluster uses a **dual-network design**: Ethernet for management and shared storage, Thunderbolt for high-speed node-to-node communication (if needed).

Network Type	Speed	Purpose
Ethernet (Gigabit)	1 Gbps	SSH management, NFS shared storage, internet access
Thunderbolt Bridge	40 Gbps (Thunderbolt 4)	Direct node-to-node communication (if needed for parallel tasks)
Wi-Fi	N/A	Not used — unreliable for cluster communication

⚠ Critical: Do not use Wi-Fi for cluster communication. Direct Thunderbolt connection between nodes is essential for performance. Using a Thunderbolt hub can introduce latency and reduce speeds. Daisy-chain directly from node to node.

5. Power & Electricity

One of the most remarkable aspects of this cluster is its **extremely low power consumption**. The Mac Mini M4 draws between **4W at idle** and **45W at full load**, with peaks up to 65W. This makes the entire 24-node cluster more efficient than a single gaming PC.

Scenario	Per Unit	24 Units Total	Daily kWh	Monthly Cost (UK)	Yearly Cost (UK)
All Idle	4W	96W	2.3 kWh	~£17.40	~£210
Mixed Load (typical workday)	~25W avg	~600W	~14.4 kWh	~£72	~£876
All Full Load (24/7 max)	45W	1,080W	~25.9 kWh	~£194	~£2,365
All Peaking (65W each)	65W	1,560W	~37.4 kWh	~£281	~£3,420

UK electricity rate used: £0.25/kWh

⚡ **Key Insight:** At full load, the entire 24-node cluster consumes **1,080W** — less than a single high-end gaming PC with an RTX 4090 (which can draw 600W+ by itself). This cluster runs on **one standard UK 13A socket** ($240V \times 13A = 3,120W$ capacity). No special electrical work required.

6. Heat & Cooling

The Mac Mini M4 generates **very little heat** compared to traditional servers or gaming PCs. Each unit dissipates between **14-154 BTU/h** depending on load. The entire cluster produces less heat than a typical gaming PC.

Scenario	Per Unit (BTU/h)	24 Units (BTU/h)	Heat Output (Watts)	Equivalent To
Idle	~14 BTU/h	~336 BTU/h	~100W	One human body
Mixed Load	~85 BTU/h	~2,040 BTU/h	~600W	Small space heater
Full Load	~154 BTU/h	~3,888 BTU/h	~1,140W	Medium space heater

🧊 **Cooling Requirements:** The cluster does **not require dedicated air conditioning**. Standard office ambient temperatures (20-25°C) are sufficient. Ensure proper rack spacing (3D printed brackets provide this) for airflow. The M4 Mac Mini's fan is silent under load — no server-grade noise.

Rack Design Note: The 3D printed rackmount stand ensures that each Mac Mini has adequate space for airflow. The stand design (from the GitHub repository) places each unit with a small gap, allowing air to circulate between machines. This prevents heat buildup and ensures consistent performance across all 24 nodes.

7. Software Stack — What Runs on Each Node

Each Mac Mini runs the same software stack, configured for its dedicated project. The stack is lightweight, containerized, and designed for persistence.

Layer	Software	Purpose
Operating System	Ubuntu 22.04 LTS (ARM64)	Lightweight, free, Docker-native, headless management
Container Runtime	Docker + Docker Compose	Isolation, portability, easy updates
Windows Compatibility	Wine 9.0	Runs Windows EXEs (games, editors, tools)
Remote Desktop	KasmVNC	Browser-accessible desktop for each app, persistent sessions
App Router	WineJS Translator (Node.js + Express)	Routes /appname to the correct KasmVNC instance
Upload Portal	DumbDrop	Drag-and-drop file upload to shared storage
Download Portal	FileServer	Password-protected file download from shared storage
Shared Storage	NFS (mounted on all nodes)	Single source of truth for assets, game files, models

Layer	Software	Purpose
Process Management	PM2	Auto-restart services, self-healing
Reverse Proxy	Nginx + Let's Encrypt SSL	HTTPS termination, routing

8. Shared Storage — The Collaborative Hub

All 24 nodes share a single NFS-mounted storage volume. This is the **collaboration hub** — any file uploaded to / uploads is immediately visible to every node in the cluster.


Upload Portal (DumbDrop)
 URL: /upload
 Drag & drop any file
 No password required (optional PIN)
 Files appear in /var/www/uploads


Download Portal (FileServer)
 URL: /download
 Password-protected
 Browse and download files from shared storage
 Works on any device


Shared Path
 /var/www/uploads
 All containers mount this as /uploads
 All nodes mount via NFS
 Single source of truth


Workflow Example
 1. Upload a model to /upload
 2. Open MilkShape on any node
 3. Load the model from /uploads
 4. Edit and save back to /uploads
 5. Download from /download

9. Projects — One Per Node

Each Mac Mini is dedicated to **one game project**. This ensures dedicated resources, no contention, and persistent state per project.

Node	Project	Original Game	Target Engine
mini01	Project IGI	Project IGI	Operation Flashpoint
mini02	Nosferatu (GUN)	Nosferatu: Wrath of Malachi	F.E.A.R.
mini03	Nosferatu (SWORD)	Nosferatu: Wrath of Malachi	Dark Messiah (Source)
mini04	The Thing	The Thing	RTCW / Doom 3 / Quake 3
mini05	Wolfschanze 1944	Wolfschanze 1944	Half-Life 2 (Source)
mini06	LEGO Space Spyrius	LEGO Space Spyrius	LEGO Racers / LEGO Island 2
mini07	Star Wars I-II-III	Star Wars Episode I-II-III	OpenJKDF2
mini08	Jedi Knight DF2 HD	Jedi Knight Dark Forces II	OpenJKDF2
mini09	IGI2	IGI2	OpenMOHAA
mini10	MOHPA: Vietnam	MOHPA: Vietnam	Medal of Honor Pacific Assault
mini11	Prince of Persia 3D	Prince of Persia 3D	Thief 1 / Thief 2
mini12	Hidden & Dangerous	Hidden & Dangerous	Operation Flashpoint

Node	Project	Original Game	Target Engine
mini13	007 Tomorrow Never Dies	007 Tomorrow Never Dies (PS1)	Hitman Codename 47 / 007 Nightfire
mini14	Blade	Blade (PS1)	Matrix: Path of Neo / Enter the Matrix
mini15	Kill.Switch	Kill.Switch	Rainbow Six Vegas 1
mini16	Transformers: The Movie	Transformers: The Movie	Transformers TAKARA
mini17	The Fifth Element	The Fifth Element	ONI
mini18	Vigilance	Vigilance (1998)	Max Payne
mini19	Terminator: Future Shock	Terminator: Future Shock	Terminator 3: War of the Machines
mini20	Starship Troopers	Starship Troopers	Serious Sam: First Encounter
mini21	Die Hard Trilogy	Die Hard Trilogy	Die Hard: Nakatomi Plaza
mini22	Vampire: Masquerade	Vampire: The Masquerade - Redemption	ArxFatalis
mini23	Spider-Man 1-2-3	Spider-Man 1-2-3	Spider-Man: The Movie
mini24	Return To Castle Wolfenstein	Return To Castle Wolfenstein	F.E.A.R.

10. Cost Breakdown

Item	Quantity	Unit Cost (Est.)	Total Cost (Est.)	Notes
Mac Mini M4 (16GB, 256GB)	24	~£600	~£14,400	Base model is sufficient
19" Server Rack	1	~£300	~£300	Second-hand available
3D Printed Brackets (STL)	24 sets	~£20 filament	~£20	Print yourself or use a service
PDU Power Strips (12-socket)	4	~£40	~£160	Each handles 6 Mac Minis
24-port Gigabit Switch	1	~£100	~£100	Managed or unmanaged
Thunderbolt Cables (1m)	23	~£25	~£575	For daisy-chaining
Ethernet Cables	24	~£5	~£120	Cat6 recommended
Rack Shelves	4	~£30	~£120	Brackets attach to these
Total Hardware Cost			~£15,795	

11. Workflow — How You Work



The Workflow:

1. **Inspiration** — Watch a gameplay video on your iPad Air 2, or see something in a film.

- 2. **Open Dashboard** — On your MacBook Pro + ViewFinity, open the FORGE dashboard HTML.
- 3. **Click Project** — Click the multi-link for the project you want to work on.
- 4. **All Tools Open** — 5-10 tabs open instantly: original game, target engine, map editor, tools, reference.
- 5. **Work** — Each tab is a live, persistent session. Everything is exactly where you left it.
- 6. **Switch** — Click another project. New window opens. All tools ready. No setup.
- 7. **Cross-Pollinate** — Learn something in one engine, apply it to another instantly.
- 8. **Compile in Background** — Compilation runs on one node while you work on another.
- 9. **Close Tab** — KasmVNC keeps the session alive. Come back later, same state.

The result: You work in a state of continuous flow. No context switching cost. No "where was I?" moments. Just creation.

12. Cluster Management

The cluster is managed from your MacBook Pro using a single command-line tool: `forge-ctl`. This script provides full control over all 24 nodes.

Command	What It Does
<code>forge-ctl status</code>	Checks all 24 nodes — online/offline, what's running
<code>forge-ctl restart</code>	Reboots all 24 nodes
<code>forge-ctl update</code>	Runs <code>apt-get update/upgrade</code> on all nodes
<code>forge-ctl docker</code>	Shows running containers on all nodes
<code>forge-ctl dashboard</code>	Opens the cluster dashboard in your browser

Remote Access: You can SSH into any node directly from your MacBook Pro: `ssh root@miniXX.local`

13. Persistence — State That Never Dies

The cluster is designed for **persistent state**. When you close a tab, the KasmVNC session **keeps running** on the node. When you reopen the tab, you're exactly where you left off.

What Persists	How It Works
KasmVNC Sessions	Container keeps running even when browser tab is closed
Wine Prefix	Persistent volume mounted inside container
Game State	Saved to shared storage or local volume
Editor State	Open files, cursor positions, unsaved changes — all preserved
Ubuntu OS	ZFS snapshots (local + NAS) for full system state

 **Cold Start:** If all 24 nodes are powered off, the cluster can be booted from snapshots. Each node restores its last state from the NAS, and all 24 are ready in ~5-10 minutes.

14. Snapshot Imaging — The Backup System

Each node's state is backed up using **ZFS snapshots**. This allows for full system restore, rollback, and cold booting from a saved state.

Component	Storage Location	Snapshot Frequency	Restore Time
OS + Base Image	NAS (master image)	Rebuilt when updating	~2 min per node
Node State (Wine + Game)	Local SSD + NAS	Daily, and before shutdown	~5 min per node
Shared Storage	NAS (RAID)	Continuous (RAID)	Instant (mount)

Snapshot Workflow:

1. End of day: snapshot-node.sh save on each node
2. Snapshot is sent to NAS as a ZFS stream
3. Next day: nodes boot from last saved snapshot
4. If a node breaks: snapshot-node.sh restore — full restore in minutes

15. Dashboard — Your Command Center

The cluster dashboard (forge-dashboard.html) provides a real-time view of all 24 nodes. It's the first thing you open when you sit down to work.

Column	What It Shows
#	Node number (01-24)
Node	Hostname (miniXX.local)
Project	The game project assigned to this node
Status	 Running /  Booting /  Offline
URL	Click to open the project in your browser

16. Comparison — Mac Mini Cluster vs Traditional Server

Aspect	24-Mac Mini Cluster	Traditional 24-GPU Server
Initial Cost	~£15,800	£30,000-£100,000+
Power Consumption (Full Load)	~1,080W	~5,000-10,000W+
Yearly Electricity (UK)	~£876 (mixed load)	£4,000-£10,000+
Noise Level	Silent (office-friendly)	Jet engine (requires server room)
Cooling Required	Room temperature	Dedicated AC / liquid cooling
Physical Footprint	1 standard rack	2-4 racks
Isolation	24 separate machines	1 shared machine
OS Flexibility	Each node independent	Single OS per server

Aspect	24-Mac Mini Cluster	Traditional 24-GPU Server
Maintenance	Replace one node at a time	Whole server offline

17. Why This Cluster Is NOT for AI/LLM

This cluster is designed specifically for **parallel game development**. It is **not** for AI training or LLM inference.

 **This cluster is for game development. The Supermicro H200 handles AI.**

The key difference is that game development workloads are **independent** — each node works on its own project without needing to communicate with others. This completely avoids the Thunderbolt bottleneck that affects AI clustering.

- **AI/LLM workloads** require massive inter-node communication (all-to-all). Thunderbolt bandwidth becomes a bottleneck.
- **Game development** is isolated per node. No communication needed. Perfect for this architecture.
- **Supermicro H200** with 8x NVIDIA H200 GPUs handles all AI workloads.

18. Summary — The Complete Package

 **What You Get:**

- **24 dedicated machines** — one per game project
- **24/7 persistence** — KasmVNC keeps state between sessions
- **Ultra-low power** — ~1,080W full load (less than a gaming PC)
- **Silent operation** — office-friendly, no server noise
- **One dashboard** — control all 24 nodes from one screen
- **Shared storage** — upload once, access everywhere
- **Snapshots** — full state backup and restore
- **One command management** — forge-ctl controls everything
- **Instant switching** — click a project, all tools open immediately
- **No context switching cost** — flow state maintained

 **This is the infrastructure for your 24-game remake studio.** One rack. 24 Mac Minis. 24 projects. One workflow. No friction. Just creation.